

Computer Architecture Operating Systems and Networks

HDSDEV_SEP24_Y2, CIC_SEPOL25

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Continuous Assessment

Weighting 50%

Release Date and Time: Monday 20th October at 9 AM

Submission Date and Time: Sunday 9th November at 23.55

Answer the appropriate Question(s) from each of Sections A, B and C

Learning Outcomes (LOs) for the Module:

- LO1 Identify and describe the relationship between each component of the computer system and how each individual component works
- LO2 Develop a conceptual understanding of different operating systems architectures
- LO3 Demonstrate competent use of the system utilities, tools and user interface to exercise, demonstrate and investigate the Operating System
- LO4 Have an understanding of the theory, concepts, principles, issues and limitations of network technologies and internet technologies and how these technologies are used for inter-computers communication.

Section A [50 marks] Please answer ALL questions, all questions carry 5 marks

Question 1 [5 Marks]

In terms of Input/Output devices, it is said that a user program wants to handle all devices in a uniform manner. This enhances simplicity & reduces the chances of error. Detail and discuss the elements of a device descriptor that can be used to provide this.

Question 2 [5 Marks]

Explain the purpose of the heat sink and highlight its purpose and role within a computer system. Give a brief explanation. Please also mention how the need for this component has evolved, through the different generations of computer.

Question 3 [5 Marks]

Explain Moore's Law in your own words and list two consequences of its impact on computer technology.

Question 4 [5 Marks]

Convert the following IP addresses to their binary equivalents. You MUST show your workings. a.

244.67.83.12

b. 28.66.12.30

c. 10.0.0.254

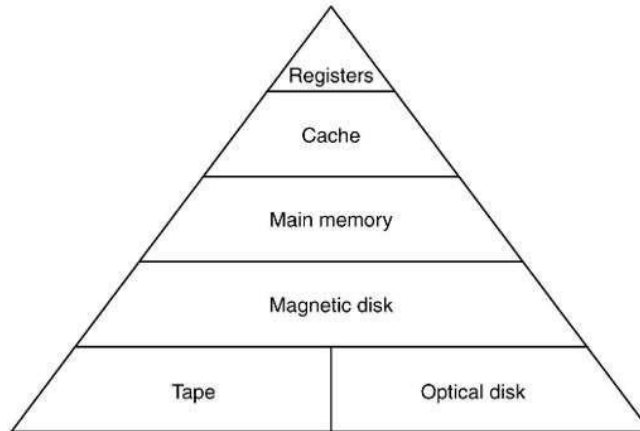
d. 172.16.5.200

e. 203.0.113.45

Question 5 [5 Marks]

Below is a diagram of the hierarchy of memory devices available for use in a computer system. Compare and contrast 2 of the layers of your choice, in terms of persistence, cost, speed etc.

Memory Hierarchies



Question 6 [5 Marks]

What are the three main responsibilities of an operating system? Give a brief explanation of each.

Question 7 [5 Marks]

Find the class of the following IP addresses.

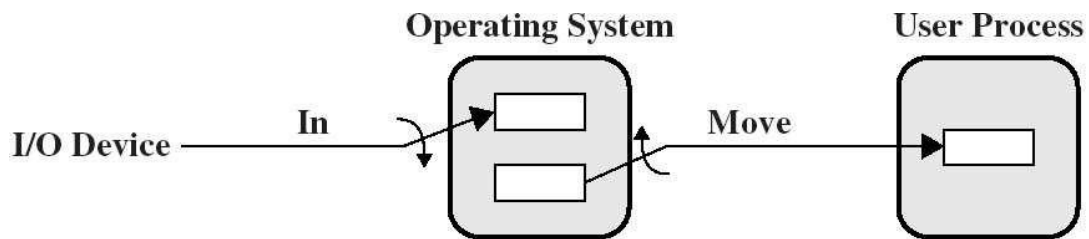
- a. 10111111 11111111 11100111 11011101
- b. 11011111 10110011 10100111 11000101
- c. 11110011 11011000 11110110 10011101
- d. 01000011 10110011 01011111 01000101

Question 8 [5 Marks]

Networks use network classes; these can be subdivided using netmasks. Please define the common subnets for Class A, B & C. how many network addresses are available to each class?

Question 9 [5 Marks]

An operating System may encounter problems in relation to the speed of various devices. Spooling and buffering are two techniques that can be used to overcome these issues. Give a brief explanation of each and discuss the benefits of using the mechanism in the diagram below.



Question 10 [5 Marks]

Compare and contrast the three categories of Users of a User Interface (UI). Why must one be careful when assigning the category of user for a given task?

Section B [20 marks]

Based on the following criteria, **answer Question 1 or Question 2** as appropriate. Both questions carry the same marks. This is a requirement for this section, if you answer the incorrect question, **you will be awarded 0 marks.**

Question 1 should be answered if the **last digit of your student ID is odd.** Please answer Question 2 if the **last digit of you student ID is even.**

Question 1 [20 Marks] Attempt this part, only if the last digit of your student id is odd.

(i) Identify and compare two non-pre-emptive scheduling algorithms [2 * 3 marks]

(ii) Consider a set of processes, which arrive at a system as follows:

Process Name	Arrival Time	Burst/ Service time (amount of time it needs to complete its processing)
P1	0	3
P2	1	7
P3	3	6
P4	8	4

By applying the First Come First Served (FCFS) algorithm, sketch out the order in which the processes would be selected for execution.

[illegible]

[8 marks]

From the resulting graph, calculate the following measures:

- (a) The average Response Time [3 marks]
- (b) The average Turnaround Time [3 marks]

Question 2 [20 Marks] Attempt this part, only if the last digit of your student id is even.

- (i) Identify and compare two non-pre-emptive scheduling algorithms [2 * 3 marks]
- (ii) Consider a set of processes, which arrive at a system as follows:

Process Name	Arrival Time	Burst/ Service time (amount of time it needs to complete its processing)
P1	0	6
P2	1	5
P3	5	6
P4	8	3

By applying the Shortest Process First (SPF) algorithm, sketch out the order in which the processes would be selected for execution.

[illegible]

P2																			
P3																			
P4																			

[8 marks]

From the resulting graph, calculate the following measures:

- (a) The average Response Time [3 marks]
- (b) The average Turnaround Time [3 marks]

Section C [30 marks]

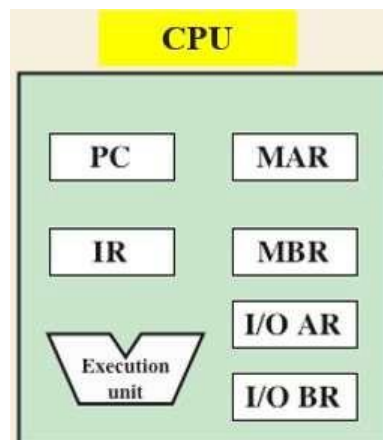
Question 1 [15 Marks]

The OSI Network Model has seven layers. Please provide a summary of the OSI model, naming and listing the responsibilities of each layer. [10 marks]

You must provide a diagram of your own, not taken from notes or online, of the OSI model. [5 marks]

Question 2 [15 Marks]

The CPU is one of the central components of a computer system. It is encapsulated in the diagram in the figure below



- i) Discuss and explain the main functions of the CPU.

[3 marks]

ii) Name and explain the main elements within the CPU: PC, IR, MAR, MBR, I/O AR & I/O BR. [6 * 2 marks]